

Landau Green's Functions and Graphene Wave Functions

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1 Nonrelativistic Landau-level Green's function

Consider a two-dimensional electron in a perpendicular magnetic field. With the Landau-gauge choice

$$\mathbf{A} = (-By, 0), \quad L = \sqrt{\frac{\hbar}{eB}}, \quad (1)$$

the Hamiltonian is

$$\hat{H}_e = \frac{\hat{\pi}^2}{2m}, \quad \hat{\pi} = \hat{p} + e\mathbf{A}. \quad (2)$$

A convenient set of eigenstates is

$$\psi_{nk_x}^e(\boldsymbol{\rho}) = \frac{e^{ik_x x}}{\sqrt{2\pi}} \varphi_n(\xi), \quad \xi = \frac{y}{L} - k_x L, \quad (3)$$

where

$$\varphi_n(\xi) = \frac{1}{\sqrt{L 2^n n! \sqrt{\pi}}} H_n(\xi) e^{-\xi^2/2}. \quad (4)$$

The corresponding energies are

$$E_n = \hbar\omega_c \left(n + \frac{1}{2} \right), \quad \omega_c = \frac{eB}{m}. \quad (5)$$

Using the convention $G_e = (H_e - E)^{-1}$, the Green's function is

$$G_e(\boldsymbol{\rho}, \boldsymbol{\rho}', E) = \sum_{n=0}^{\infty} \int_{-\infty}^{\infty} \frac{e^{ik_x(x-x')} \varphi_n(\xi) \varphi_n(\xi')}{2\pi[\hbar\omega_c(n+1/2) - E]} dk_x. \quad (6)$$

The needed Hermite integral is

$$\int_{-\infty}^{\infty} e^{-x^2} H_m(x+y) H_n(x+z) dx = 2^n \sqrt{\pi} m! z^{n-m} L_m^{n-m}(-2yz), \quad m \leq n. \quad (7)$$

After the k_x integral one obtains

$$G_e(\boldsymbol{\rho}, \boldsymbol{\rho}', E) = \frac{e^{-R^2/2+i\chi}}{2\pi\hbar\omega_c L^2} \sum_{n=0}^{\infty} \frac{L_n(R^2)}{n+1/2-\mathcal{E}}, \quad \mathcal{E} = \frac{E}{\hbar\omega_c}, \quad (8)$$

where

$$R^2 = \frac{|\boldsymbol{\rho} - \boldsymbol{\rho}'|^2}{2L^2}, \quad \chi = \frac{(x-x')(y+y')}{2L^2}. \quad (9)$$

The phase χ is gauge dependent, while the decay and pole structure are gauge invariant. The Laguerre sum may be evaluated using

$$t^{-\alpha} \sum_{n=0}^{\infty} \frac{L_n^{-\alpha}(t)}{n+a-\alpha} = \Gamma(a-\alpha) \Psi(a, \alpha+1; t), \quad (10)$$

where Ψ is the Tricomi confluent hypergeometric function. Equivalently,

$$\Psi(a, c; t) = e^{t/2} t^{-1/2-\mu} W_{\kappa, \mu}(t), \quad \kappa = \frac{c}{2} - a, \quad \mu = \frac{c}{2} - \frac{1}{2}. \quad (11)$$

Setting $a = 1/2 - \mathcal{E}$, $c = 1$, and $t = R^2$ gives

$$\boxed{G_e(\boldsymbol{\rho}, \boldsymbol{\rho}', E) = \frac{e^{i\chi}}{2\pi\hbar\omega_c L^2 |R|} \Gamma\left(\frac{1}{2} - \mathcal{E}\right) W_{\mathcal{E},0}(R^2)}. \quad (12)$$

Equivalently,

$$e^{-R^2/2} \sum_{n=0}^{\infty} \frac{L_n(R^2)}{n+1/2-\mathcal{E}} = \frac{1}{|R|} \Gamma\left(\frac{1}{2} - \mathcal{E}\right) W_{\mathcal{E},0}(R^2). \quad (13)$$

2 Monolayer graphene Landau-level Green's function

At one valley of monolayer graphene, the low-energy Hamiltonian is

$$\hat{H}_M = u(\sigma_x \hat{\pi}_x + \sigma_y \hat{\pi}_y), \quad (14)$$

where $u \simeq 10^8$ cm/s. In the same Landau gauge, the eigenstates can be written as

$$\psi_{nk_x s}^M(\boldsymbol{\rho}) = \frac{e^{ik_x x}}{\sqrt{2\pi(2-\delta_{n0})}} \begin{pmatrix} -s\varphi_{n-1}(\xi) \\ \varphi_n(\xi) \end{pmatrix}, \quad s = \pm 1, \quad (15)$$

with the convention $\varphi_{-1} = 0$. The energies are

$$E_{ns} = s\hbar\omega\sqrt{n}, \quad \omega = \frac{\sqrt{2}u}{L}. \quad (16)$$

The stationary Green's function is a 2×2 matrix. It is useful to introduce

$$\bar{E} = \frac{E}{\hbar\omega}, \quad r_{10} = \frac{(y' - y) - i(x - x')}{L\sqrt{2}}, \quad r_{01} = \frac{(y - y') - i(x - x')}{L\sqrt{2}}. \quad (17)$$

After summing over the band index s and doing the k_x integral, the diagonal components reduce to Laguerre sums,

$$G_{\sigma\sigma}^M(\boldsymbol{\rho}, \boldsymbol{\rho}', E) = \frac{\bar{E} e^{-R^2/2+i\chi}}{2\pi\hbar\omega L^2} \sum_{n=0}^{\infty} \frac{L_n(R^2)}{n+\sigma-\bar{E}^2}, \quad \sigma = 0, 1. \quad (18)$$

Using the Whittaker representation above gives

$$\boxed{G_{00}^M(\boldsymbol{\rho}, \boldsymbol{\rho}', E)} = \frac{\bar{E} e^{i\chi}}{2\pi\hbar\omega L^2 |R|} \Gamma(-\bar{E}^2) W_{\bar{E}^2+1/2,0}(R^2), \quad (19)$$

$$\boxed{G_{11}^M(\boldsymbol{\rho}, \boldsymbol{\rho}', E)} = \frac{\bar{E} e^{i\chi}}{2\pi\hbar\omega L^2 |R|} \Gamma(1-\bar{E}^2) W_{\bar{E}^2-1/2,0}(R^2). \quad (20)$$

For the off-diagonal components one uses

$$L_{n-1}^1(R^2) = \frac{n}{R^2} [L_{n-1}(R^2) - L_n(R^2)] \quad (21)$$

to obtain the compact relation

$$\boxed{G_{\sigma,1-\sigma}^M(\boldsymbol{\rho}, \boldsymbol{\rho}', E) = \frac{r_{\sigma,1-\sigma} \bar{E}}{R^2} (G_{11}^M - G_{00}^M).} \quad (22)$$

The poles occur at

$$\bar{E} = 0, \pm 1, \pm\sqrt{2}, \pm\sqrt{3}, \dots, \quad (23)$$

which reproduces the monolayer graphene Landau spectrum.

A useful local expansion of the Whittaker function is

$$W_{\lambda,0}(z) = \frac{\sqrt{z} e^{-z/2}}{\Gamma(1/2 - \lambda)^2} \sum_{k=0}^{\infty} \frac{\Gamma(k - \lambda + 1/2)}{(k!)^2} z^k [2\psi(k+1) - \psi(k - \lambda + 1/2) - \ln z]. \quad (24)$$

3 Born approximation

For a weak scalar potential $\tilde{V}(\mathbf{r})$, the Born form of the scattering wave function is

$$\Psi(\mathbf{r}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{ikx} + \int G^M(\mathbf{r}, \mathbf{r}'; E) \tilde{V}(\mathbf{r}') \Psi(\mathbf{r}') d^2 r'. \quad (25)$$

This formula is the starting point for weak-potential scattering from a graphene dot, impurity, or electrostatic island in a magnetic field.

4 Massive Dirac wave functions in symmetric gauge

Now consider the massive Dirac Hamiltonian

$$H = v_F \boldsymbol{\sigma} \cdot (-i\hbar \boldsymbol{\nabla} - e\mathbf{A}) + \Delta \sigma_z, \quad \mathbf{A} = \frac{Br}{2} \hat{\boldsymbol{\theta}}, \quad (26)$$

with magnetic length $l_B = \sqrt{\hbar/(eB)}$. In polar coordinates,

$$\sigma_r = \begin{pmatrix} 0 & e^{-i\theta} \\ e^{i\theta} & 0 \end{pmatrix}, \quad \sigma_\theta = \begin{pmatrix} 0 & -ie^{-i\theta} \\ ie^{i\theta} & 0 \end{pmatrix}. \quad (27)$$

With these conventions the Hamiltonian is

$$H = \begin{pmatrix} \Delta & -i\hbar v_F e^{-i\theta} \left(\partial_r - \frac{i}{r} \partial_\theta - \frac{r}{2l_B^2} \right) \\ -i\hbar v_F e^{i\theta} \left(\partial_r + \frac{i}{r} \partial_\theta + \frac{r}{2l_B^2} \right) & -\Delta \end{pmatrix}. \quad (28)$$

The conserved total angular momentum is

$$J_z = -i\hbar \partial_\theta + \frac{\hbar}{2} \sigma_z. \quad (29)$$

An eigenstate of J_z may be written as

$$\Phi(r, \theta) = \begin{pmatrix} \Phi_A(r)e^{im\theta} \\ i\Phi_B(r)e^{i(m+1)\theta} \end{pmatrix}, \quad m \in \mathbb{Z}. \quad (30)$$

The Dirac equation $H\Phi = E\Phi$ gives the coupled radial equations

$$\left(\partial_r + \frac{r}{2l_B^2} - \frac{m}{r} \right) \Phi_A(r) = -\frac{E + \Delta}{\hbar v_F} \Phi_B(r), \quad (31)$$

$$\left(\partial_r + \frac{m+1}{r} - \frac{r}{2l_B^2} \right) \Phi_B(r) = \frac{E - \Delta}{\hbar v_F} \Phi_A(r). \quad (32)$$

Eliminating Φ_B gives

$$\left[\partial_r^2 + \frac{1}{r} \partial_r + \frac{m+1}{l_B^2} - \frac{r^2}{4l_B^4} - \frac{m^2}{r^2} + q^2 \right] \Phi_A(r) = 0, \quad q^2 = \frac{E^2 - \Delta^2}{(\hbar v_F)^2}. \quad (33)$$

Normalizable solutions decay as $e^{-r^2/(4l_B^2)}$ at large radius and behave as $r^{|m|}$ near the origin. Set

$$\eta = \frac{r^2}{2l_B^2}. \quad (34)$$

For $m \geq 0$,

$$\Phi_A^+(r) = \mathcal{N}_+ r^m e^{-r^2/(4l_B^2)} {}_1F_1 \left(-\frac{l_B^2 q^2}{2}, m+1, \frac{r^2}{2l_B^2} \right). \quad (35)$$

For $m < 0$,

$$\Phi_A^-(r) = \mathcal{N}_- r^{|m|} e^{-r^2/(4l_B^2)} {}_1F_1 \left(-m - \frac{l_B^2 q^2}{2}, 1-m, \frac{r^2}{2l_B^2} \right). \quad (36)$$

The lower component follows from the first-order equation,

$$\Phi_B(r) = -\frac{\hbar v_F}{E + \Delta} \left(\partial_r + \frac{r}{2l_B^2} - \frac{m}{r} \right) \Phi_A(r). \quad (37)$$

For $m \geq 0$, this gives explicitly

$$\Phi_B^+(r) = \mathcal{N}_+ \frac{E - \Delta}{2\hbar v_F(m+1)} r^{m+1} e^{-r^2/(4l_B^2)} {}_1F_1 \left(1 - \frac{l_B^2 q^2}{2}, m+2, \frac{r^2}{2l_B^2} \right). \quad (38)$$

For $m < 0$, if $a = -m - l_B^2 q^2/2$ and $b = 1 - m$, then

$$\Phi_B^-(r) = -\mathcal{N}_- \frac{\hbar v_F}{E + \Delta} r^{|m|-1} e^{-r^2/(4l_B^2)} \left[2|m| {}_1F_1(a, b, \eta) + \frac{a}{b} \frac{r^2}{l_B^2} {}_1F_1(a+1, b+1, \eta) \right]. \quad (39)$$

The normalization constants $\mathcal{N}_\pm$ are fixed by the boundary conditions of the scattering or bound-state problem.

5 Partial-wave expansion for circular scattering

For a circular graphene quantum dot or circular electrostatic island, angular momentum diagonalizes the scattering problem. An incident massless Dirac plane wave moving along x can be expanded as

$$\Phi_k^i(r, \theta) = \frac{1}{\sqrt{2}} e^{ikr \cos \theta} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \sum_{m=-\infty}^{\infty} i^m \begin{pmatrix} J_m(kr) e^{im\theta} \\ iJ_{m+1}(kr) e^{i(m+1)\theta} \end{pmatrix}. \quad (40)$$

The outgoing reflected wave is expanded in Hankel functions,

$$\Phi_k^r(r, \theta) = \frac{1}{\sqrt{2}} \sum_{m=-\infty}^{\infty} a_m^r i^m \begin{pmatrix} H_m^{(1)}(kr) e^{im\theta} \\ iH_{m+1}^{(1)}(kr) e^{i(m+1)\theta} \end{pmatrix}. \quad (41)$$

The coefficients a_m^r are determined by matching the spinor wave functions at the radius of the circular scatterer.